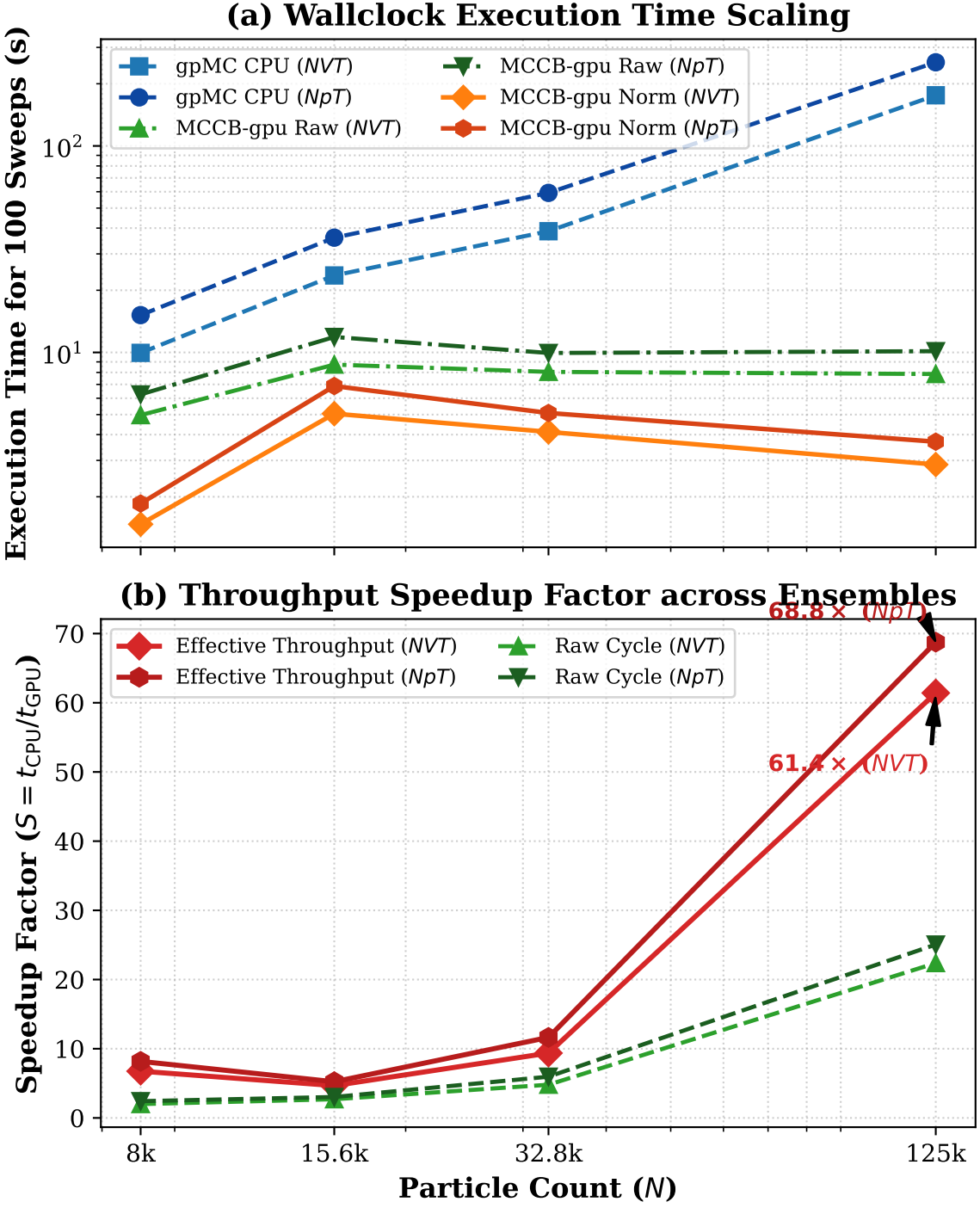
(a) Wallclock Execution Time Scaling
Execution Time for 100 Sweeps (s)
gpMC CPU (NVT)
gpMC CPU (NpT)
MCCB-gpu Raw (NVT)
MCCB-gpu Raw (NpT)
MCCB-gpu Norm (NVT)
MCCB-gpu Norm (NpT)
10^2
10^1
(b) Throughput Speedup Factor across Ensembles
Speedup Factor ($S = t_{CPU}/t_{GPU}$)
Effective Throughput (NVT)
Effective Throughput (NpT)
Raw Cycle (NVT)
Raw Cycle (NpT)
68.8× (NpT)
61.4× (NVT)
0
10
20
30
40
50
60
70
8k
15.6k
32.8k
125k
Particle Count (N)